

reports, and detecting fraudulent transactions by recognising linguistic anomalies. Education technology platforms employ LLMs for personalised tutoring, content generation, and automated grading, thereby enhancing both teaching effectiveness and student engagement. Beyond these, manufacturing, legal services, and creative industries all exploit the adaptability of open source LLMs to streamline processes, generate insights, and unlock new business opportunities.

Open source vs commercial LLMs: A comparison

A fundamental distinction between open source and commercial LLMs lies in accessibility. Open source models are freely available for download, modification, and deployment, whereas commercial offerings often impose usage restrictions and recurring costs. This openness translates to greater customisation potential, enabling organisations to tailor models to domain-specific vocabularies or

compliance requirements. In contrast, commercial LLMs typically provide limited transparency regarding their internal workings and training data, raising concerns about explainability and potential biases. However, commercial models may offer dedicated support, regular updates, and optimised performance out of the box. Ultimately, the choice between open source and commercial LLMs hinges on an organisation's priorities—whether they value autonomy and adaptability or prioritise convenience and vendor support (see Table 1).

Fine-tuning open source LLMs: Steps and resources

Fine-tuning is the process of adapting a pre-trained open source LLM to excel in a particular task or domain. The journey begins with data collection, where practitioners curate high-quality, representative datasets relevant to their objectives. The next step involves preprocessing and formatting this data to align with the model's input

requirements. Fine-tuning is conducted using frameworks such as Hugging Face Transformers or PyTorch, leveraging robust computational resources—often GPUs or TPUs—to update the model's weights. It is crucial to monitor training metrics, adjust hyperparameters, and apply early stopping to prevent overfitting. Upon completion, the fine-tuned model should undergo rigorous evaluation against validation datasets to ensure it meets performance standards. Documentation and version control throughout the process are essential for reproducibility and future enhancements.

Validating reliability and ensuring data protection

Reliability and data protection are paramount when deploying open source LLMs in sensitive environments. To validate reliability, practitioners should employ comprehensive evaluation protocols, including cross-validation, adversarial testing, and benchmarking against established datasets. Regular audits help detect and rectify unintended

Table 1: A comparison of open source and commercial LLMs

Aspect	Open source LLM	Commercial LLM
Accessibility	Freely available for download and local deployment, with the flexibility to modify and adapt as required.	Usually accessible through paid subscriptions or APIs, often with restrictions on usage and modification.
Customisation	Can be extensively customised to suit specific needs, including domain-specific vocabulary and compliance requirements.	Customisation options are generally limited, with most models offering standardised performance out of the box.
Transparency	Source code, model architecture, and training data are typically open for review, promoting explainability and trust.	Internal workings and training datasets are often proprietary, making it difficult to assess or address potential biases.
Support	Relies on community-driven forums and documentation for troubleshooting and updates.	Usually backed by dedicated customer support, regular updates, and performance optimisations by the vendor.
Cost	Minimal to no licensing fees, lowering the barrier for adoption by startups, academics, and enterprises.	Involves recurring costs, such as subscription fees or pay-per-use charges, which can increase with scale.
Innovation pace	Benefits from rapid advancements and knowledge sharing within a global open source community.	Improvements are governed by the provider's release cycles and commercial priorities.
Security and compliance	Allows for greater control over data handling and compliance with specific regulatory requirements.	May offer built-in compliance features, but users have limited visibility or control over internal processes.